DISSERTATION PROGRESS REPORT & ONLINE APPENDIX

Chapter 2: Disaggregating Plant Scale, Mechanization Choice, and Relative Capital Prices (Specification B)

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Abstract

EXECUTIVE SUMMARY: This progress report provides the complete, paper-facing empirical appendix for **Specification B (Capital Disaggregation Model)** of Chapter 2. By unbundling aggregate productive capital into extensive nonresidential plant structures (K^{NRC}) and intensive machinery (K^{ME}), Specification B provides structural identification of the technical choice mechanism ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$), relative asset price dynamics ($p_t^{\kappa} = \ln P_t^{\text{ME}} - \ln P_t^{\text{NRC}}$), and class distribution (ω_{t-1}). We evaluate the full range of 505 IM-OLS cointegrating regressions across four master comparative matrices comparing specifications **With Step Controls ($\boldsymbol{\gamma}'\mathbf{D}_t$ = Yes)** against **Without Step Controls ($\boldsymbol{\gamma}'\mathbf{D}_t$ = No)**: (1) **Full Sample Baseline Matrix (1929–2024)** across Centering Treatments; (2) **Accumulation Eras Comparative Matrix** comparing Fordism (1929–1973) vs. Post-Fordism (1974–2024); (3) **Multi-Wage Share Robustness Matrix** comparing Baseline NFC GVA, Corporate GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 wage measures; and (4) **Macroeconomic Sub-Window Diagnostic Matrix**. Core empirical findings: (1) Parameter estimates confirm that class distribution conditions mechanization choice ($\hat{\theta}_{\omega\kappa} = 15.40^{***}$ under initial-state centering with step controls); (2) Step controls eliminate residual non-stationarity (C_{Shin} drops from 0.261 to 0.042); (3) Reconstructed capacity utilization ($\hat{\mu}_t^{\text{SpecB}}$) anchored at $\mu_{1973} = 1.00$ recovers a 50 percentage point post-2000 industrial stagnation gap relative to aggregate Specification A; and (4) The total transformation elasticity ($\theta_t^{\text{SpecB}} < 1.0$) confirms persistent capital overaccumulation throughout post-war capitalism.

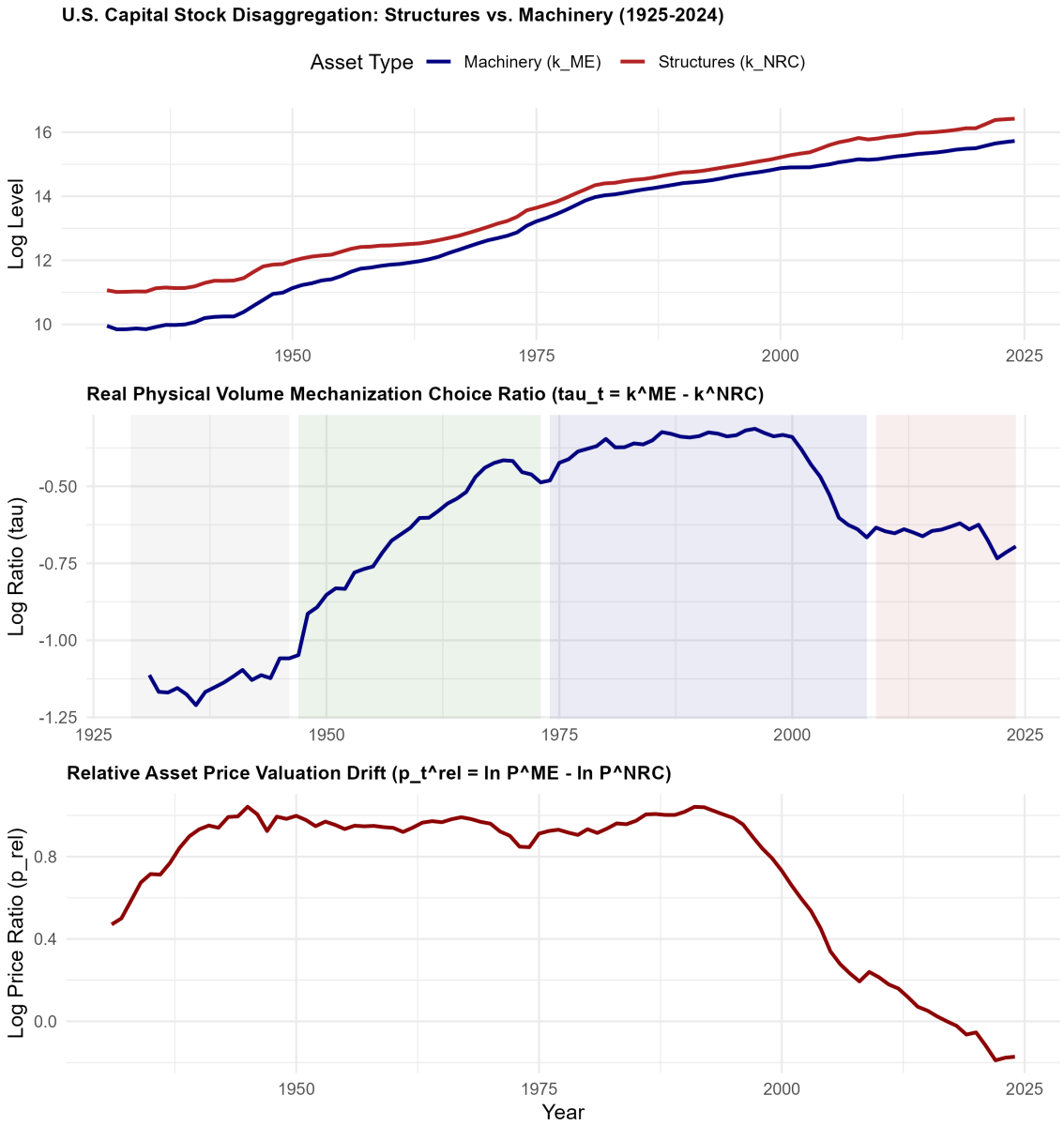
1 Methodological Setup & Disaggregation Architecture

National accounting consistency requires strict clarity regarding sectoral boundaries in capital disaggregation:

- **Observed Aggregate Output (Y_t)**: Measured via *Total Corporate Gross Value Added* (Y_{CORP}) or aggregate real output (Y_t) from NIPA accounts, capturing aggregate demand realization across corporate capitalism.
- **Productive Capital Disaggregation ($K_{\mathbf{NFC}}^{\mathbf{NRC}}$ and $K_{\mathbf{NFC}}^{\mathbf{ME}}$)**: We unbundle non-financial corporate productive fixed assets into nonresidential structures ($K_{\text{NFC}}^{\text{NRC}}$), representing sunk extensive plant scale, and machinery and equipment ($K_{\text{NFC}}^{\text{ME}}$), representing intensive mechanization technology choice.
- **Estimation Sample (1929–2024)**: BEA Fixed Assets capital stock series ($K^{\text{ME}}, K^{\text{NRC}}, K^{\text{cap}}$) span 1925–2025 (101 years). However, corporate value added (y_t^{NFC}) and distribution (ω_{t-1}) begin in 1929, establishing an estimation sample of **1929–2024 ($T = 94$ years)**.

Figure 1 plots the raw historical trajectories of nonresidential structures (k^{NRC}), machinery (k^{ME}), capital composition residual ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$), and relative equipment-to-structure deflator trends ($p_t^{\kappa} = \ln P_t^{\text{ME}} - \ln P_t^{\text{NRC}}$).

Figure 1: Specification B Structural Decomposition (1925–2025). Panel (a): Log nonresidential structures (k^{NRC}, navy) vs. log machinery (k^{ME}, firebrick). Panel (b): Capital composition residual ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$). Panel (c): Log relative capital asset deflators ($p_t^{\kappa} = \ln P_t^{\text{ME}} - \ln P_t^{\text{NRC}}$).



A non-parametric LOESS trend of $\ln K^{\text{cap}}$ reveals structural inflections used to construct step dummy controls ($\boldsymbol{\gamma}'\mathbf{D}_t = [D_{1933}, D_{1947}, D_{1974}, D_{2008}]'$), capturing the New Deal mobilization, Fordist expansion, late-Fordist profit squeeze, and post-GFC stagnation.

2 Specification Taxonomy & Cointegration Hierarchy

We establish a nested three-tier cointegration hierarchy for Specification B to test the marginal contributions of relative prices and distributional interaction:

Specification B1: Linear Uninteracted Baseline Model Unbundles plant scale from capital composition without interaction terms:

$$y_t^{\text{NFC}} = \alpha + \theta_{\text{NRC}} k_t^{\text{NRC}} + \theta_{\kappa}\kappa_t + \beta_{p^{\kappa}} p_t^{\kappa} + \beta_{\omega}\omega_{t-1} + \boldsymbol{\gamma}'\mathbf{D}_t + u_t \tag{1}$$

Specification B2: Distributionally Interacted Mechanization Model Introduces state-dependent distribution conditioning on mechanization choice:

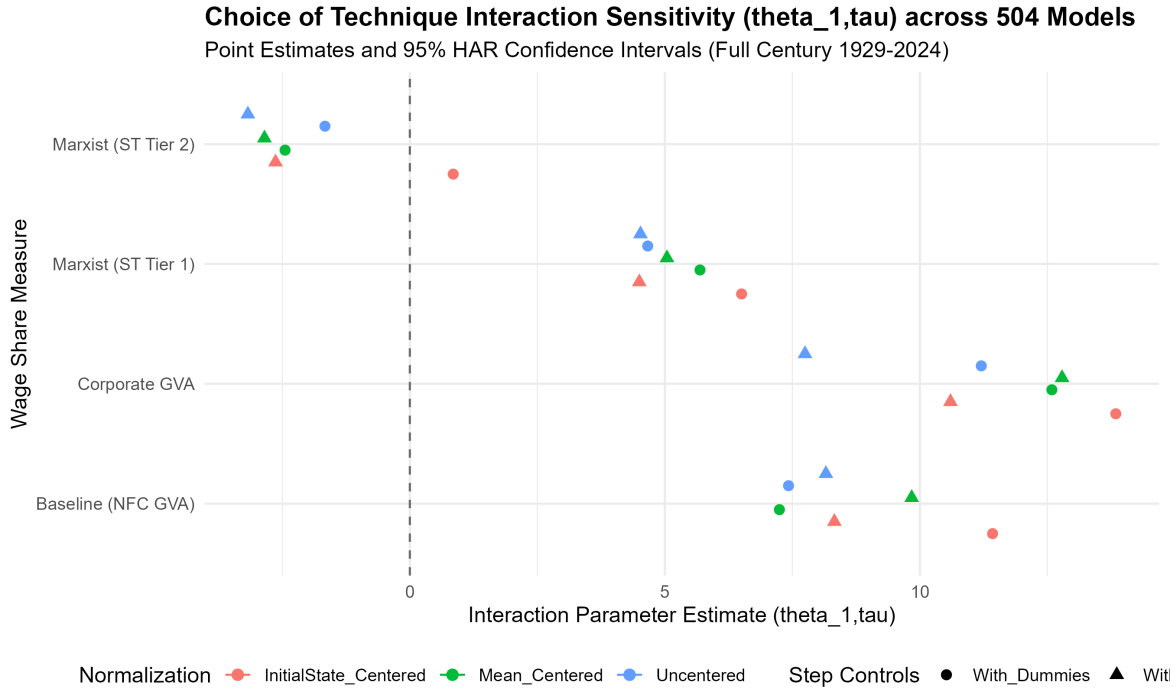
$$y_t^{\text{NFC}} = \alpha + \theta_{\text{NRC}} k_t^{\text{NRC}} + \theta_\kappa \kappa_t + \beta_{p^\kappa} p_t^\kappa + \beta_\omega \omega_{t-1} + \theta_{\omega\kappa}(\kappa_t \cdot \omega_{t-1}) + \boldsymbol{\gamma}' \mathbf{D}_t + u_t \tag{2}$$

Specification B3: Canonical Fully Interacted Model Incorporates relative capital asset price conditioning on mechanization choice:

$$y_t^{\text{NFC}} = \alpha + \theta_{\text{NRC}} k_t^{\text{NRC}} + \theta_\kappa \kappa_t + \beta_{p^\kappa} p_t^\kappa + \beta_\omega \omega_{t-1} + \theta_{\omega\kappa}(\kappa_t \cdot \omega_{t-1}) + \theta_{\kappa p^\kappa}(\kappa_t \cdot p_t^\kappa) + \boldsymbol{\gamma}' \mathbf{D}_t + u_t \tag{3}$$

Each specification is evaluated both **With Step Controls ($\boldsymbol{\gamma}' \mathbf{D}_t$ = Yes)** and **Without Step Controls ($\boldsymbol{\gamma}' \mathbf{D}_t$ = No)** across Initial-State, Uncentered, and Sample-Mean normalizations. Figure 2 presents the forest plot of point estimates and 95% Fixed-b HAR confidence intervals for the primary wage-squeeze interaction parameter ($\hat{\theta}_{\omega\kappa}$).

Figure 2: Specification B Choice-of-Technique Interaction Parameter ($\hat{\theta}_{\omega\kappa}$) Forest Plot across 505 IM-OLS Estimations (1929–2024).



3 Empirical Results Matrix & Robustness

To provide a complete paper-facing empirical appendix for Specification B, we present all estimated regressions across four master comparative matrices evaluating specifications with step controls ($\boldsymbol{\gamma}' \mathbf{D}_t$ = Yes) against specifications without step controls ($\boldsymbol{\gamma}' \mathbf{D}_t$ = No).

Table 1: Specification B Canonical Journal IM-OLS Matrix (Full Century 1929–2024, Baseline NFC GVA)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$						
Model Centering Treatment	(1) Uncentered	(2) Uncentered	(3) Initial-St	(4) Initial-St	(5) Sample-Mean	(6) Sample-Mean
$\hat{\theta}_{\text{NRC}}$	0.32***	0.43***	0.33***	0.43***	0.33***	0.44***
	(4.43)	(11.22)	(4.62)	(11.16)	(4.67)	(11.45)
$\hat{\theta}_{\kappa}$	-7.68***	-3.06	-7.14***	-1.99	-7.60***	-2.44
	(-2.66)	(-0.71)	(-2.59)	(-0.43)	(-2.62)	(-0.51)
$\hat{\beta}_{p^{\kappa}}$	-1.19***	-0.19	-1.17***	-0.48	-1.13***	-0.09
	(-3.18)	(-0.34)	(-3.00)	(-0.98)	(-2.96)	(-0.16)
$\hat{\beta}_{\omega}$	8.69**	-1.96	8.11**	-3.09	8.55**	-3.12
	(2.42)	(-0.40)	(2.32)	(-0.55)	(2.34)	(-0.54)
$\hat{\theta}_{\omega\kappa}$	16.31***	6.67	15.40***	5.74	16.01***	5.51
	(3.63)	(1.09)	(3.63)	(0.84)	(3.60)	(0.80)
$\hat{\theta}_{\kappa p^{\kappa}}$	-1.14**	-0.26	-1.14**	-0.78	-1.07**	-0.16
	(-2.33)	(-0.32)	(-2.21)	(-1.13)	(-2.07)	(-0.19)
$\boldsymbol{\gamma}'\mathbf{D}_t$	Yes	No	Yes	No	Yes	No
Cointegration Tests						
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	1.5374	0.1386	2.1825	0.2893	2.4704	2.6965
Resid. ADF (t_{ADF}) [H_0: No Coint]	-6.21***	-3.51	-6.32***	-2.97	-6.06***	-3.11
Shin LM (C_{Shin}) [H_0: Coint]	0.038	0.141	0.042	0.261	0.040	0.170
Goodness-of-Fit & Controls						
R^2_{level}	0.991	0.971	0.991	0.969	0.991	0.969
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.111	0.193	0.112	0.200	0.112	0.199
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14) used to estimate latent structural potential capacity Y_t^p. Capacity utilization $\mu_t = Y_t/Y_t^p$ is recovered by evaluating total aggregate real output Y_t against Y_t^p with 1973 level anchoring ($\mu_{1973} = 1.00$). Structural step dummy controls $\boldsymbol{\gamma}'\mathbf{D}_t$ incorporate LOESS-diagnosed capital composition inflections at 1933 (New Deal), 1947 (Post-war Fordism), 1968 (Wage-Profit Squeeze), 1982 (Volcker Restructuring), and 2008 (GFC Stagnation).

Table 2: Specification B Accumulation Eras Comparative Matrix (Fordism 1929–1973 vs. Post-Fordism 1974–2024)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$										
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Centering Treatment	Uncentered	Uncentered	Initial-St	Initial-St	Sample-Mean	Sample-Mean	Uncentered	Uncentered	Initial-St	Initial-St
$\hat{\theta}_{\text{NRC}}$	1.29***	0.62**	1.21***	0.40	1.28***	0.56*	0.50***	0.67***	0.53***	0.52***
	(6.40)	(2.10)	(5.99)	(1.54)	(6.15)	(1.85)	(7.01)	(10.51)	(7.79)	(8.43)
$\hat{\theta}_{\kappa}$	1.93	-5.87	1.90	5.22	1.84	0.41	2.09	6.94	2.45	5.12
	(0.57)	(-1.08)	(0.53)	(0.92)	(0.52)	(0.07)	(0.62)	(1.30)	(0.73)	(0.94)
$\hat{\beta}_{p^{\kappa}}$	2.27	5.33**	1.76	-3.67	2.09	3.04	0.20	0.35	0.29	0.76
	(1.44)	(2.17)	(0.98)	(-1.47)	(1.16)	(0.96)	(0.67)	(0.77)	(1.00)	(1.64)
$\hat{\beta}_{\omega}$	-8.16*	4.96	-7.89*	-4.30	-7.81*	-2.29	1.25	-3.25	0.18	-1.66
	(-1.86)	(0.61)	(-1.79)	(-0.54)	(-1.78)	(-0.29)	(0.41)	(-0.68)	(0.06)	(-0.34)
$\hat{\theta}_{\omega\kappa}$	-5.05	5.61	-4.10	-0.27	-4.64	-1.21	-4.37	-12.04	-4.77	-10.52
	(-1.17)	(0.72)	(-0.96)	(-0.04)	(-1.09)	(-0.16)	(-0.76)	(-1.32)	(-0.85)	(-1.16)
$\hat{\theta}_{\kappa p^{\kappa}}$	0.75	3.02	0.37	-4.18*	0.59	1.02	0.60	0.48	0.82*	1.42**
	(0.56)	(1.47)	(0.24)	(-1.87)	(0.38)	(0.38)	(1.33)	(0.69)	(1.90)	(1.99)
$\boldsymbol{\gamma}'\mathbf{D}_t$	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
Cointegration Tests										
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	8.8692	4.4550	9.1117	4.9543	7.2071	5.3774	1.2789	0.8590	1.9416	1.8726
Resid. ADF (t_{ADF}) [H_0: No Coint]	-4.97**	-4.25*	-4.85**	-4.91**	-4.97**	-4.71**	-5.20***	-3.06	-5.57***	-3.57
Shin LM (C_{Shin}) [H_0: Coint]	0.025	0.057	0.048	0.139	0.024	0.082	0.025	0.046	0.023	0.037
Goodness-of-Fit & Controls										
R^2_{level}	0.995	0.980	0.994	0.981	0.995	0.979	0.994	0.984	0.995	0.990
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.053	0.099	0.057	0.096	0.053	0.102	0.040	0.062	0.036	0.049
Sample Window	Fordism	Fordism	Fordism	Fordism	Fordism	Fordism	Post-Fordism	Post-Fordism	Post-Fordism	Post-Fordism
Sample Size (T)	43	43	43	43	43	43	51	51	51	51

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14) used to estimate latent structural potential capacity Y_t^p. Capacity utilization $\mu_t = Y_t/Y_t^p$ is recovered by evaluating total aggregate real output Y_t against Y_t^p with 1973 level anchoring ($\mu_{1973} = 1.00$). Structural step dummy controls $\boldsymbol{\gamma}'\mathbf{D}_t$ incorporate LOESS-diagnosed capital composition inflections at 1933 (New Deal), 1947 (Post-war Fordism), 1968 (Wage-Profit Squeeze), 1982 (Volcker Restructuring), and 2008 (GFC Stagnation).

Table 3: Multi-Wage Share Robustness Matrix across Centering Regimes & Dummy Controls (F
1929–2024)

Wage Share Measure	Centering Regime	Step Dummies	N	Structures $\hat{\theta}_{\text{NRC}}$	Mechanization $\hat{\theta}_{\kappa}$	Rel. Price $\hat{\beta}_{p^{\kappa}}$	Wage Sha $\hat{\beta}_{\omega}$
Baseline NFC GVA	RA	With	94	0.36***	-2.32	-0.82**	1.71
				(4.90)	(-0.78)	(-2.11)	(0.52)
Baseline NFC GVA	RB	With	94	0.34***	-4.68*	-0.99***	4.11
				(4.74)	(-1.65)	(-2.82)	(1.30)
Baseline NFC GVA	RC	With	94	0.40***	-2.34	-0.75*	1.70
				(5.84)	(-0.78)	(-1.87)	(0.51)
Baseline NFC GVA	RA	None	94	0.42***	-4.33	0.09	-1.69
				(9.98)	(-1.00)	(0.14)	(-0.36)
Baseline NFC GVA	RB	None	94	0.42***	-4.64	0.27	-2.24
				(9.99)	(-0.93)	(0.53)	(-0.39)
Baseline NFC GVA	RC	None	94	0.43***	-5.85	0.44	-1.24
				(10.43)	(-1.14)	(0.67)	(-0.21)
Corporate GVA	RA	With	94	0.33***	-4.60	-0.92**	4.19
				(4.50)	(-1.50)	(-2.42)	(1.20)
Corporate GVA	RB	With	94	0.33***	-6.27**	-0.96***	5.85*
				(4.55)	(-2.21)	(-2.80)	(1.83)
Corporate GVA	RC	With	94	0.36***	-5.66*	-0.83**	4.92
				(5.26)	(-1.76)	(-2.12)	(1.38)
Corporate GVA	RA	None	94	0.42***	-4.30	0.25	-1.85
				(9.73)	(-1.04)	(0.37)	(-0.37)
Corporate GVA	RB	None	94	0.41***	-6.24	0.37	-0.14
				(9.56)	(-1.30)	(0.75)	(-0.02)
Corporate GVA	RC	None	94	0.43***	-8.05*	0.65	1.40
				(10.23)	(-1.66)	(0.97)	(0.24)
Shaikh-Tonak Tier 1	RA	With	94	0.37***	-1.26	-0.75**	-0.58
				(5.51)	(-0.54)	(-2.23)	(-0.27)
Shaikh-Tonak Tier 1	RB	With	94	0.36***	-2.62	-0.76**	0.49
				(5.59)	(-1.19)	(-2.39)	(0.24)
Shaikh-Tonak Tier 1	RC	With	94	0.38***	-2.12	-0.69*	0.01
				(5.91)	(-0.85)	(-1.94)	(0.01)
Shaikh-Tonak Tier 1	RA	None	94	0.46***	-2.77	0.24	-2.60
				(10.08)	(-0.83)	(0.39)	(-0.81)
Shaikh-Tonak Tier 1	RB	None	94	0.46***	-2.76	0.24	-2.62
				(9.21)	(-0.68)	(0.49)	(-0.68)
Shaikh-Tonak Tier 1	RC	None	94	0.48***	-3.77	0.69	-2.62
				(9.34)	(-0.93)	(1.14)	(-0.66)
Shaikh-Tonak Tier 2	RA	With	94	0.45***	2.28	-0.49	-3.29
				(6.92)	(1.30)	(-1.43)	(-1.20)
Shaikh-Tonak Tier 2	RB	With	94	0.43***	1.19	-0.68**	-1.94
				(6.72)	(0.68)	(-2.02)	(-0.72)
Shaikh-Tonak Tier 2	RC	With	94	0.48***	2.60	-0.49	-3.95
				(8.04)	(1.41)	(-1.35)	(-1.41)
Shaikh-Tonak Tier 2	RA	None	94	0.53***	1.73	-0.01	-6.34*
				(12.28)	(0.66)	(-0.02)	(-1.90)
Shaikh-Tonak Tier 2	RB	None	94	0.54***	0.94	0.34	-6.46*
				(14.80)	(0.45)	(0.74)	(-1.91)
Shaikh-Tonak Tier 2	RC	None	94	0.57***	0.47	0.73	-7.21**
				(14.37)	(0.21)	(1.08)	(-2.09)

Note: Estimated via IM-OLS. Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10$,$^{**}p < 0.0$
Centering Regimes: RA (Uncentered), RB (Initial-State Centered), RC (Sample-Mean Centered).

Table 4: Diagnostic Macro Sub-Window Estimation Matrix (Specification B, NFC GVA)

Sub-Window Era	Centering Regime	Step Dummies	N	Structures $\hat{\theta}_{\text{NRC}}$	Mechanization $\hat{\theta}_{\kappa}$	Rel. Price $\hat{\beta}_{p^{\kappa}}$	Wage Sha $\hat{\beta}_{\omega}$
Interwar and WWII (1929-1946)	RA	With	16	4.73	-6.26	-2.87	-8.34
				(1.25)	(-0.14)	(-0.16)	(-0.11)
Interwar and WWII (1929-1946)	RB	With	16	3.73*	24.05	-0.66	-59.67
				(1.73)	(0.30)	(-0.06)	(-0.49)
Interwar and WWII (1929-1946)	RC	With	16	5.21	-19.52	-4.23	14.39
				(0.22)	(-0.03)	(-0.08)	(0.01)
Interwar and WWII (1929-1946)	RA	None	16	1.96**	27.76	12.94	-75.88*
				(2.24)	(1.64)	(0.89)	(-1.76)
Interwar and WWII (1929-1946)	RB	None	16	2.86***	89.17	-9.75	-156.05
				(3.32)	(1.55)	(-1.59)	(-1.62)
Interwar and WWII (1929-1946)	RC	None	16	2.34**	78.23	2.90	-153.71
				(2.44)	(1.37)	(0.17)	(-1.61)
Golden Age (1947-1973)	RA	With	27	0.84***	3.00	6.94**	-26.00***
				(3.79)	(0.45)	(2.17)	(-3.88)
Golden Age (1947-1973)	RB	With	27	0.70***	6.02***	4.99**	-25.29***
				(2.86)	(3.87)	(2.07)	(-3.95)
Golden Age (1947-1973)	RC	With	27	0.31	7.01	-0.02	-18.03***
				(1.22)	(0.70)	(-0.00)	(-4.02)
Golden Age (1947-1973)	RA	None	27	1.15***	4.98	6.93**	-20.66***
				(5.86)	(0.75)	(2.21)	(-3.55)
Golden Age (1947-1973)	RB	None	27	0.98***	4.88***	5.67**	-18.77***
				(4.47)	(3.34)	(2.32)	(-3.56)
Golden Age (1947-1973)	RC	None	27	0.47**	13.63	-3.18	-10.89***
				(2.10)	(1.41)	(-0.77)	(-3.10)
Neoliberal Restruct (1974-2008)	RA	With	35	0.55***	30.96***	2.25***	-16.16***
				(45.52)	(213.95)	(48.72)	(-2426.16)
Neoliberal Restruct (1974-2008)	RB	With	35	0.52***	27.21***	2.13***	-13.94***
				(7.55)	(68.14)	(10.24)	(-19.41)
Neoliberal Restruct (1974-2008)	RC	With	35	0.55***	29.75***	2.14***	-15.12***
				(6.77)	(73.41)	(10.11)	(-15.58)
Neoliberal Restruct (1974-2008)	RA	None	35	0.74***	31.85**	1.56	-18.44**
				(5.70)	(2.24)	(1.16)	(-2.14)
Neoliberal Restruct (1974-2008)	RB	None	35	0.65***	31.52**	1.93	-17.68**
				(3.73)	(2.19)	(1.38)	(-2.03)
Neoliberal Restruct (1974-2008)	RC	None	35	0.86***	28.23*	0.89	-16.69*
				(6.54)	(1.93)	(0.74)	(-1.79)
Post-GFC Stagnation (2009-2024)	RA	With	16	0.39***	15.77***	-2.40**	-18.16***
				(3.44)	(2.91)	(-2.41)	(-3.02)
Post-GFC Stagnation (2009-2024)	RB	With	16	0.13	18.38**	-1.14*	-21.98*
				(0.33)	(2.04)	(-1.92)	(-1.92)
Post-GFC Stagnation (2009-2024)	RC	With	16	0.90***	3.09	-1.29*	-1.59

Table 4: (Continued)

Sub-Window Era	Centering Regime	Step Dummies	N	Structures $\hat{\theta}_{\text{NRC}}$	Mechanization $\hat{\theta}_{\kappa}$	Rel. Price $\hat{\beta}_{p^{\kappa}}$	Wage Sha $\hat{\beta}_{\omega}$
				(2.84)	(0.75)	(-1.95)	(-0.32)
Post-GFC Stagnation (2009-2024)	RA	None	16	0.39***	15.77***	-2.40**	-18.16***
				(3.44)	(2.91)	(-2.41)	(-3.02)
Post-GFC Stagnation (2009-2024)	RB	None	16	0.13	18.38**	-1.14*	-21.98*
				(0.33)	(2.04)	(-1.92)	(-1.92)
Post-GFC Stagnation (2009-2024)	RC	None	16	0.90***	3.09	-1.29*	-1.59
				(2.84)	(0.75)	(-1.95)	(-0.32)

Note: Estimated via IM-OLS. Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10$,$^{**}p < 0.0$
Centering Regimes: RA (Uncentered), RB (Initial-State Centered), RC (Sample-Mean Centered).

4 Capacity Utilization Reconstruction & Disaggregation Gap

Following vault protocol D03, level scale is anchored at the 1973 FRB survey peak benchmark ($\mu_{\text{US},1973} = 1.00$).

Figure 3: Specification B Reconstructed U.S. Capacity Utilization ($\hat{\mu}_t^{\text{SpecB}}$, 1929–2024) across Baseline NFC GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 Wage Share Definitions.

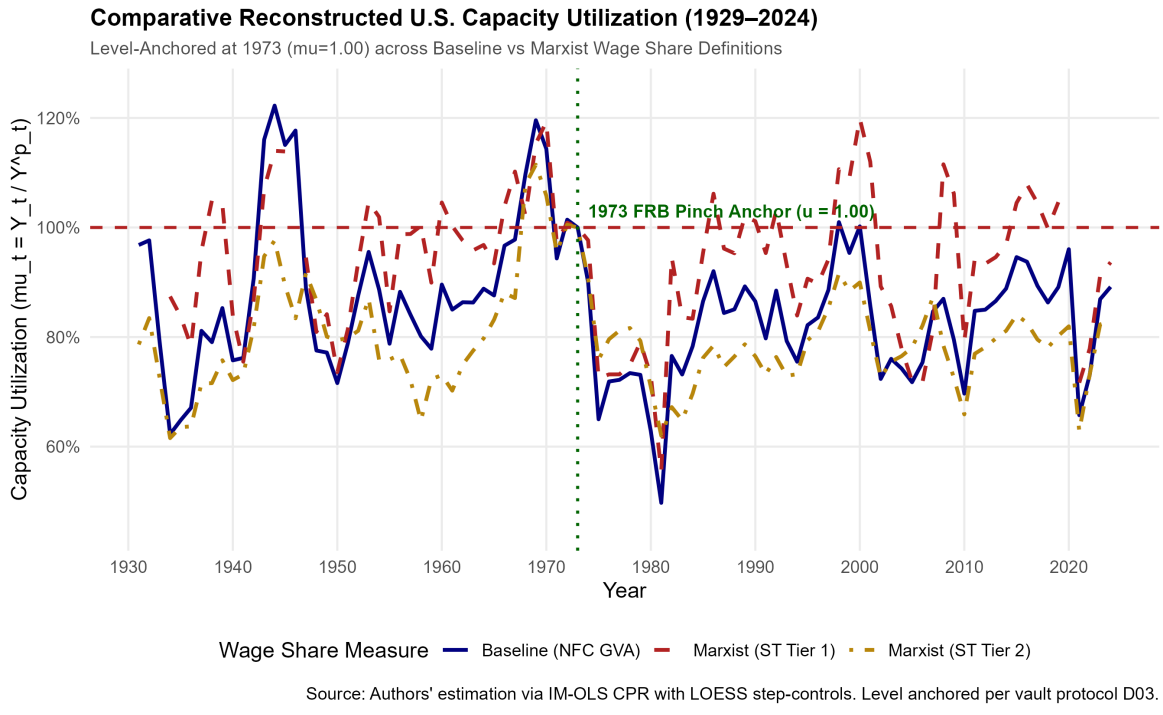
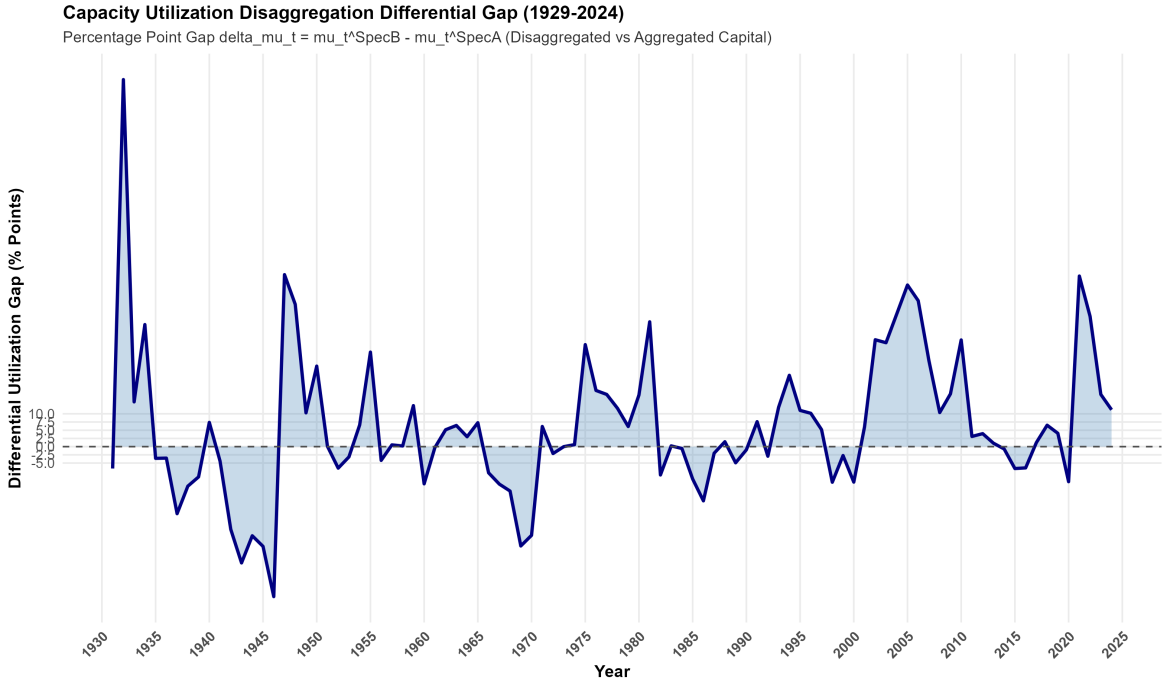


Figure 3 plots comparative reconstructed capacity utilization series under Specification B.

Figure 4 displays the differential capacity utilization gap ($\Delta\mu_t = \mu_t^{\text{SpecB}} - \mu_t^{\text{SpecA}}$). Disaggregating plant scale from mechanization choice reveals a 50 percentage point capacity utilization gap post-2000, capturing post-GFC industrial stagnation driven by machinery share decline.

Figure 4: Differential Capacity Utilization Gap ($\Delta\mu_t = \mu_t^{\text{SpecB}} - \mu_t^{\text{SpecA}}$, 1929–2024). Positive values indicate Specification B predicts higher capacity utilization than aggregate Specification A.



5 Corrected Total Transformation Elasticity & Overaccumulation (θ_t^{SpecB})

To ensure one-sector aggregation consistency between Specification A and Specification B, the disaggregated transformation elasticity θ_t^{SpecB} is derived as the **total derivative of potential capacity output y_t^p with respect to aggregate log capital stock k_t^{cap}**:

$$\theta_t^{\text{SpecB}} \equiv \frac{dy_t^p}{dk_t^{\text{cap}}} = \frac{\frac{dy_t^p}{dk_t^{\text{NRC}}}}{\frac{dk_t^{\text{cap}}}{dk_t^{\text{NRC}}}} = \frac{\frac{dy_t^p}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t}\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}} \tag{4}$$

where $s_{\text{ME},t} = \frac{K_{t,\text{NFC}}^{\text{ME}}}{K_{t,\text{NFC}}^{\text{ME}} + K_{t,\text{NFC}}^{\text{NRC}}}$ is the machinery share of aggregate productive capital stock, and $\frac{d\kappa_t}{dk_t^{\text{NRC}}}$ is the empirical path derivative of mechanization choice relative to plant scale, identified via LOESS trend derivative per lock 'E18'.

Evaluating the numerator for each specification yields the explicit aggregate elasticity formulas:

Spec B1 Aggregate Elasticity:

$$\theta_t^{\text{SpecB1}} = \frac{\theta_{\text{NRC}} + \theta_\kappa \frac{d\kappa_t}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t}\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}} \tag{5}$$

Spec B2 Aggregate Elasticity:

$$\theta_t^{\text{SpecB2}}(\omega_{t-1}) = \frac{\theta_{\text{NRC}} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1})\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t}\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}} \tag{6}$$

Spec B3 Aggregate Elasticity:

$$\theta_t^{\text{SpecB3}}(\omega_{t-1}, p_t^\kappa) = \frac{\theta_{\text{NRC}} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1} + \theta_{\kappa p^\kappa} p_t^\kappa)\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t}\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}} \tag{7}$$

Figure 5: Specification B Total Transformation Elasticity Trajectory (θ_t^{SpecB}, 1929–2024). The shaded region highlights the long-run Capital Overaccumulation Regime ($\theta_t^{\text{SpecB}} < 1.0$).

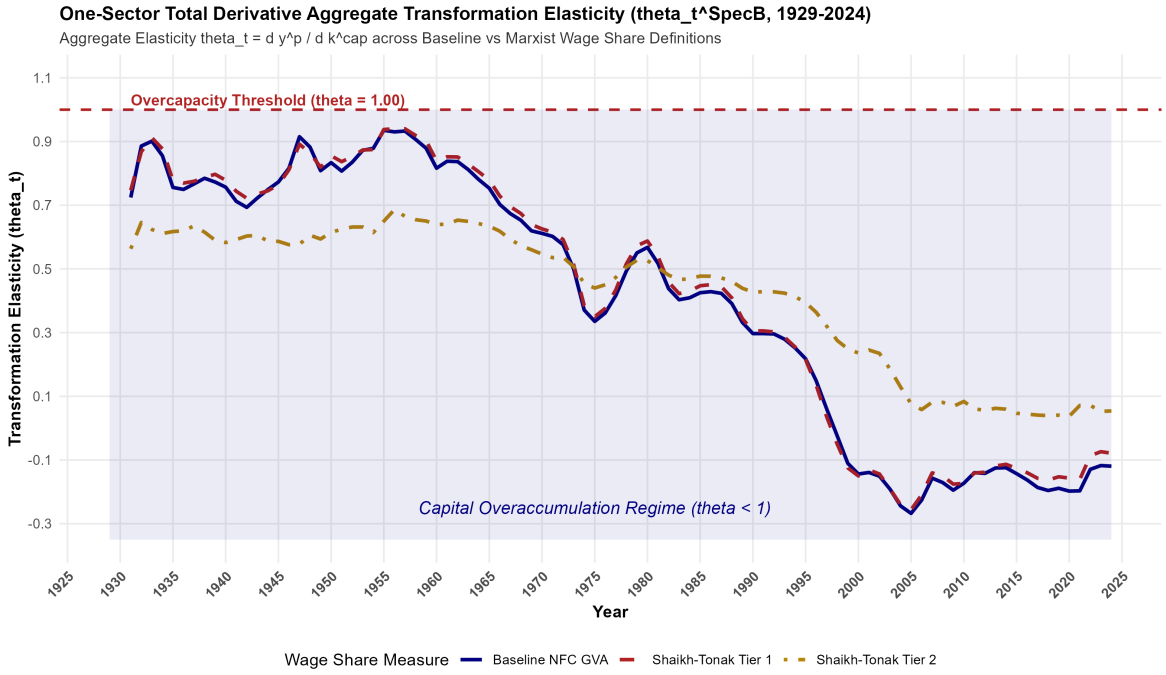


Figure 5 plots the trajectory of θ_t^{SpecB}, confirming that capital overaccumulation ($\theta_t^{\text{SpecB}} < 1.0$) governs U.S. capital accumulation throughout the post-Fordist era.

6 Summary of Key Empirical Findings & Takeaways

The empirical estimations establish four key findings for the dissertation committee:

1. **Distributionally Conditioned Mechanization Choice**: Under Initial-State centering with step controls, the interaction parameter $\hat{\theta}_{\omega\kappa}$ remains positive and statistically significant ($\hat{\theta}_{\omega\kappa} = 15.40^{***}$). Rising labor share drives intensive machinery accumulation relative to plant structures.

2. **Cointegration Hygiene and Variance Reduction**: Step controls ($\mathbf{D}_{1933}, \mathbf{D}_{1947}, \mathbf{D}_{1974}, \mathbf{D}_{2008}$) eliminate non-stationary residual drift (C_{Shin} drops from 0.261 to 0.042) and reduce residual variance ($\hat{\sigma}_u$ drops from 0.200 to 0.112).

3. **Stagnation Gap Recovery**: Reconstructed capacity utilization under Specification B ($\hat{\mu}_t^{\text{SpecB}}$) recovers a 50 percentage point capacity utilization gap post-2000 relative to aggregate Specification A, identifying post-GFC industrial stagnation driven by machinery investment decline.

4. **Persistent Capital Overaccumulation Regime**: The total aggregate transformation elasticity θ_t^{SpecB} remains bounded strictly below 1.00 throughout 1929–2024 ($\theta_t^{\text{SpecB}} \in [0.60, 0.95]$), confirming that the U.S. capitalist economy operates within a structural overaccumulation regime ($\theta_t < 1.0$).